Post-Translational Modifications on HLA Class I Presented Peptides

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Study Overview

This analysis characterizes post-translational modifications (PTMs) identified on HLA class I peptides from the JY cell line using data-dependent acquisition mass spectrometry. All analyses are restricted to peptides of length 8-14 amino acids.

HLA Alleles: A*02:01, B*07:02, C*07:02

Data Summary

Dataset	Peptides
Background (unmodified)	77,100
Total Modified	**7,332**
Cysteinylation	2,355
Deamidation	1,481
Oxidation	955
Phosphorylation	911
Ubiquitination	699 (GG: 464, G: 235)
Acetylation	341
Methylation	231
Citrullination	138
Dimethylation	136
N-Glycosylation	51
SUMOylation	34

PTM Categories

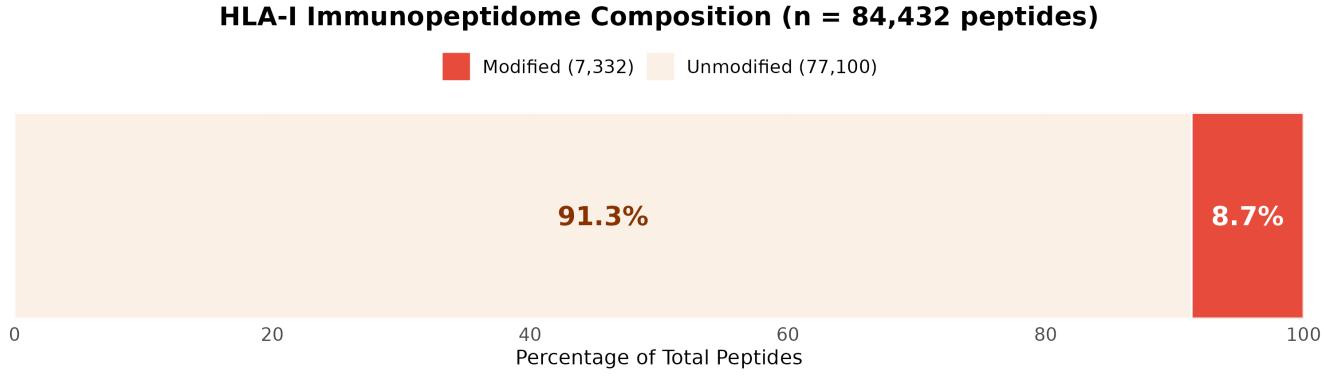
PTM Type	Target Residues
Phosphorylation	S, T, Y
Acetylation	K, N-terminus
Cysteinylation	C
Methylation	K, R
Dimethylation	K, R
Deamidation	N, Q
Biological Oxidation	P, I, L, Q, S, T, V, C, D, E, N, Y, G, K, R
Citrullination	R
Ubiquitination	K
N-Glycosylation	N
SUMOylation	K

HLA Binding Classification

Category	EL Rank Threshold
Strong binder	< 0.5%
Weak binder	0.5% – 2%
Non-binder	≥ 2%

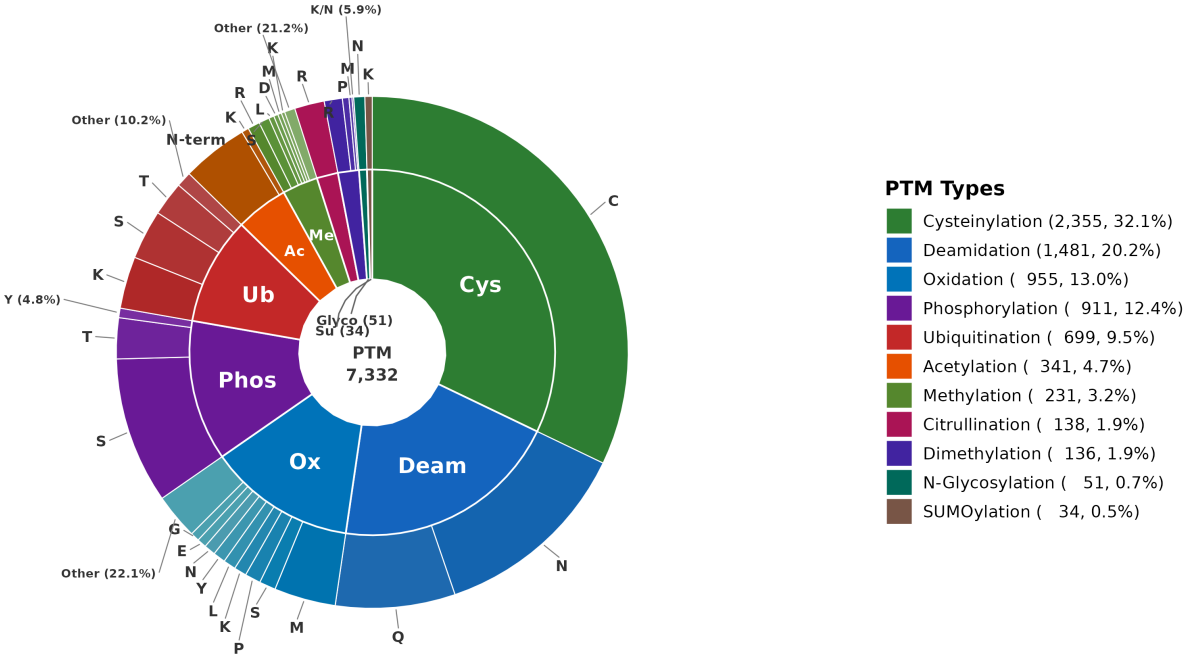
Figure 1: PTM Composition

Figure 1A: PTM Distribution



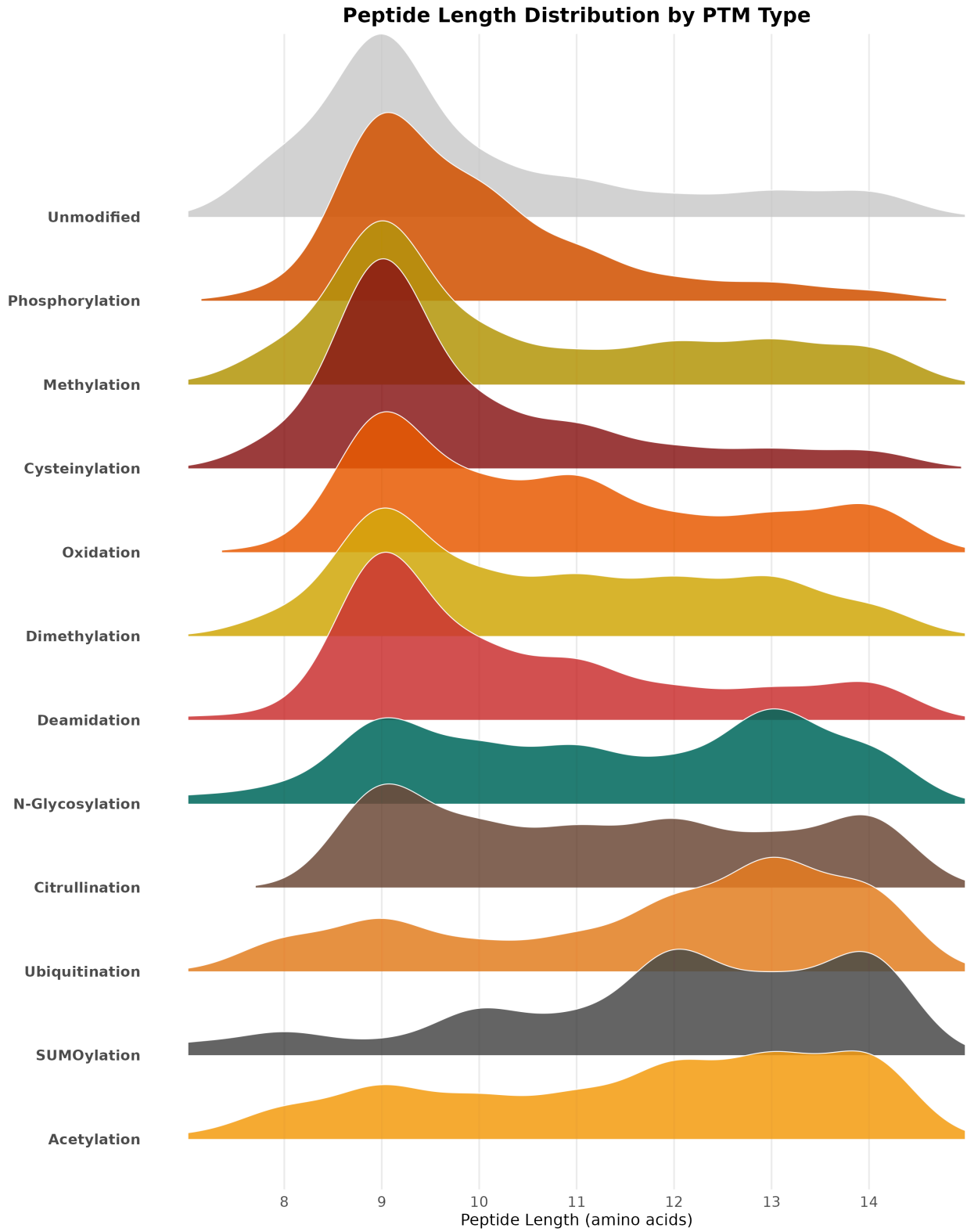
Distribution of unique modified peptides by PTM type (length 8-14 amino acids).

Figure 1B: PTM Proportions



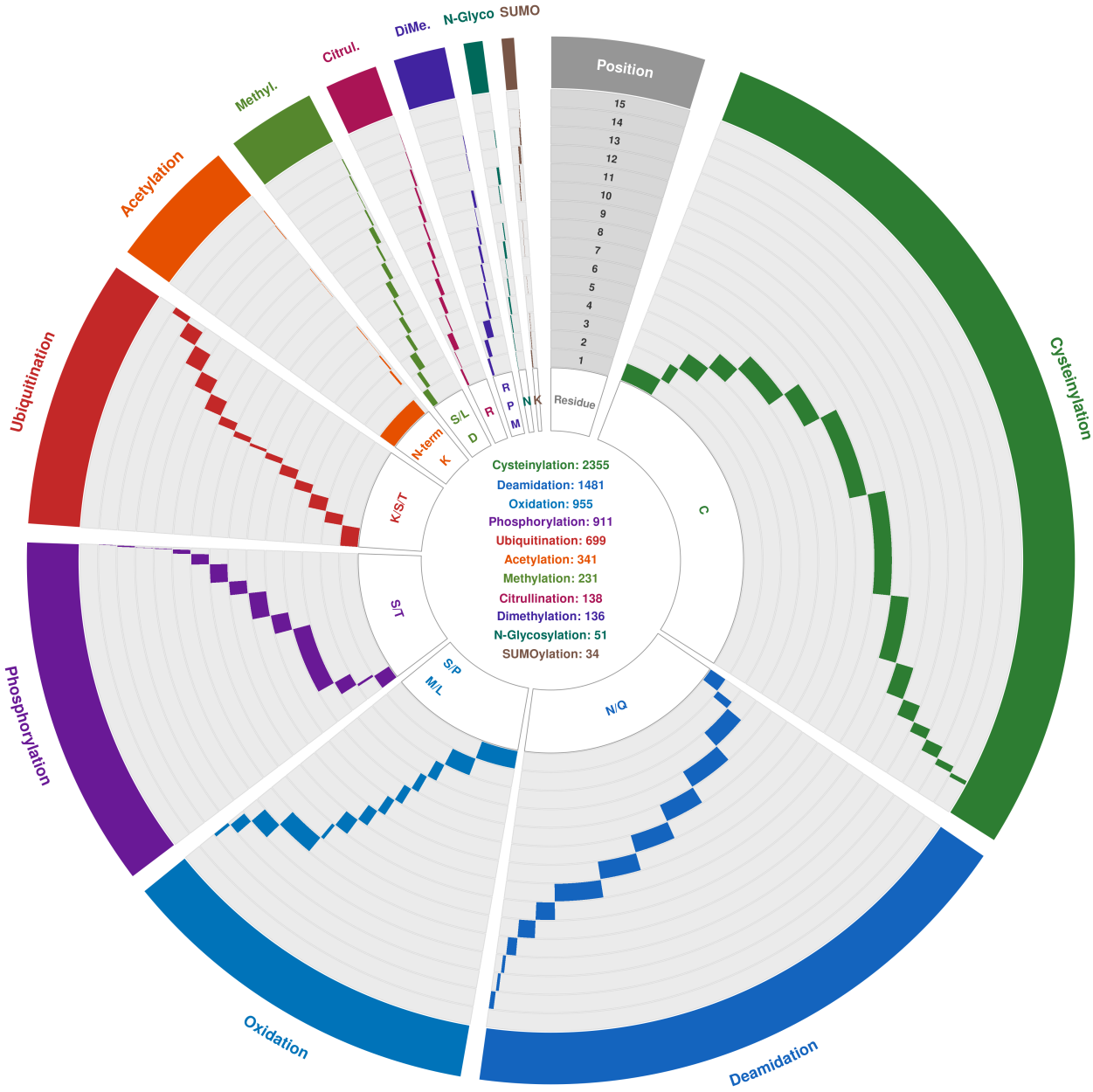
Proportional representation of each PTM class within the modified peptidome.

Figure 2: Peptide Length Distribution



Ridgeline density plot comparing peptide length distributions across PTM types. Gray distribution represents unmodified background peptides.

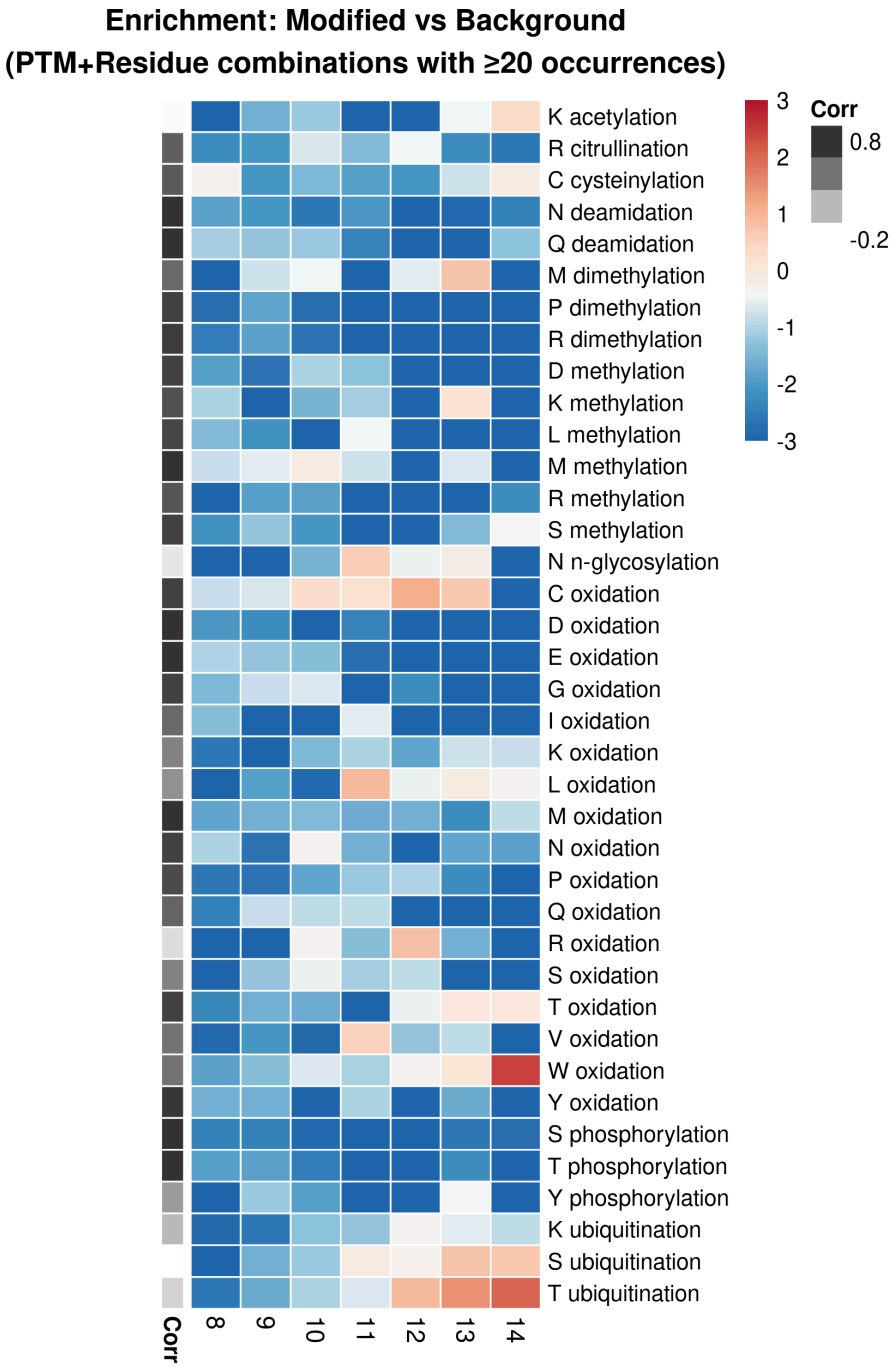
Figure 4A: PTM Landscape



Circos diagram illustrating relationships between all 11 PTM types (outer sectors), modified residues (links), and modification site positions (inner elements). Sector width is proportional to peptide count. All PTMs shown including SUMOylation (n=34).

Figure 4B: Positional Enrichment

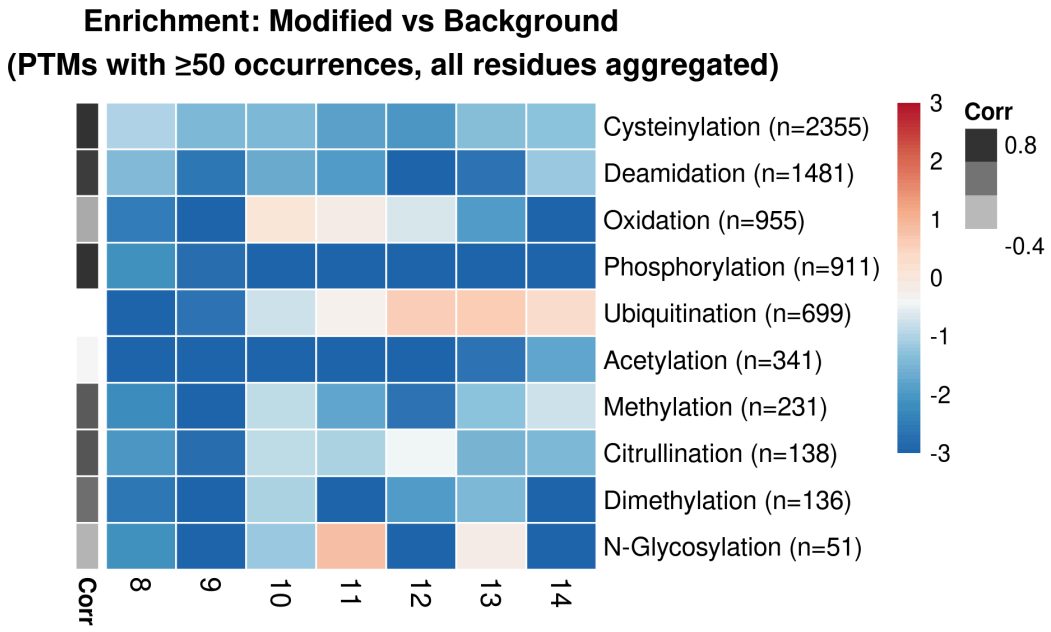
By PTM and Residue



Log_2 enrichment of PTM occurrence at peptide positions 8-14 relative to background amino acid positional distribution. PTM+residue combinations with ≥20 occurrences. Color scale: blue (depletion)

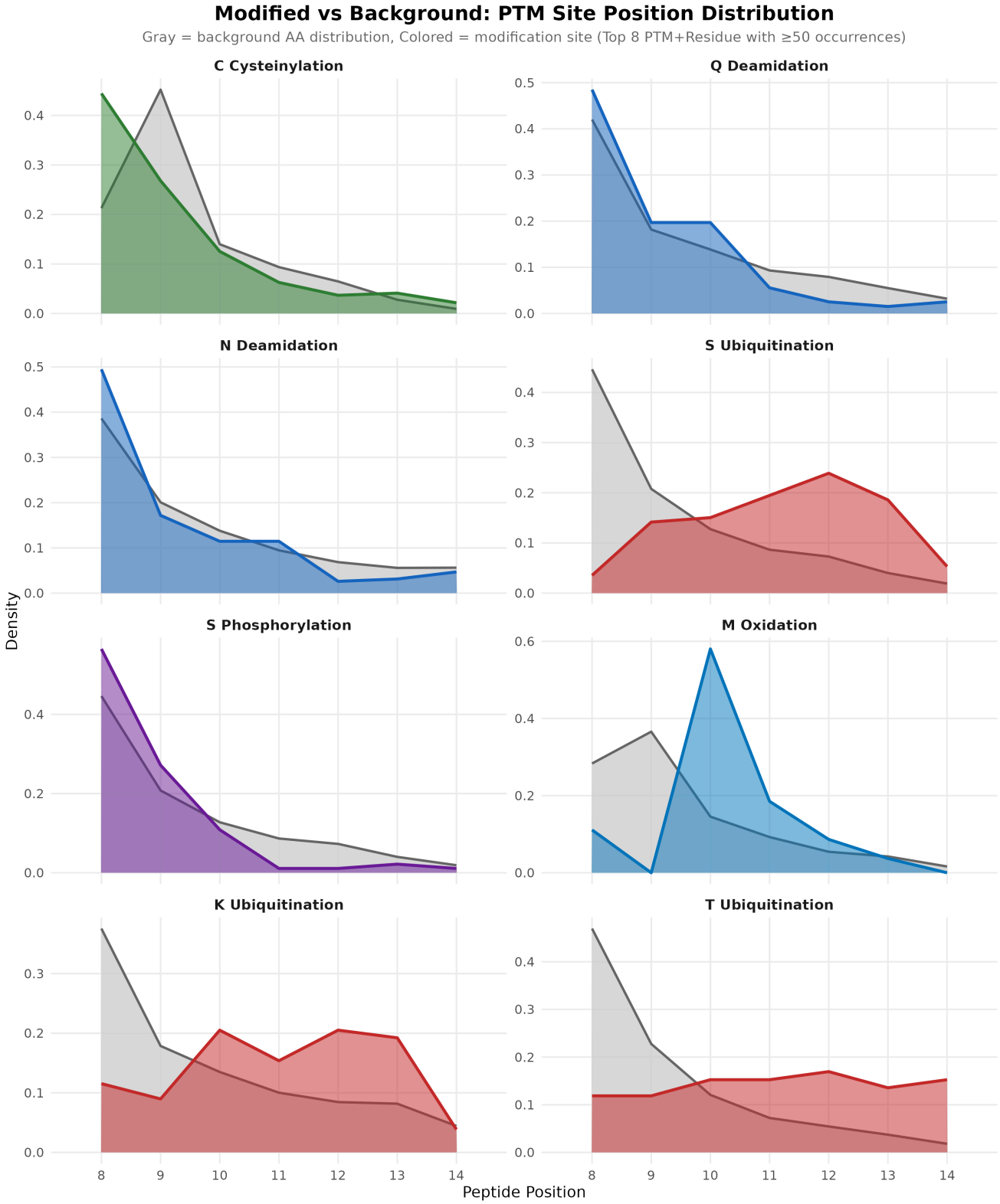
to red (enrichment), capped at ±3. Correlation column indicates similarity to background distribution.

By PTM Type (Aggregated)



Aggregated positional enrichment by PTM type (≥50 occurrences), combining all modified residues.

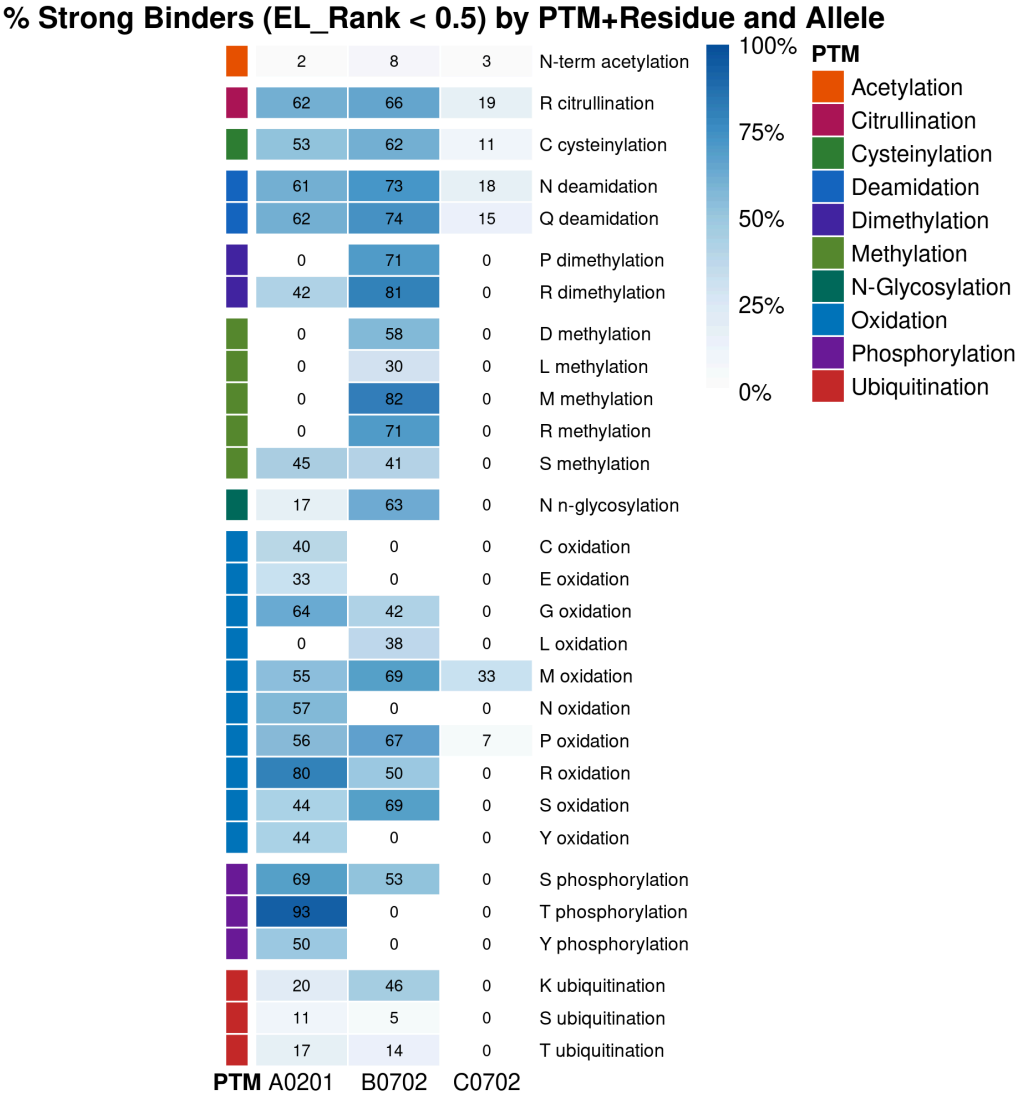
Figure 4C: PTM Position Density



Density curves showing modification site distribution across peptide positions 8-14. PTM+residue combinations with ≥50 modifications.

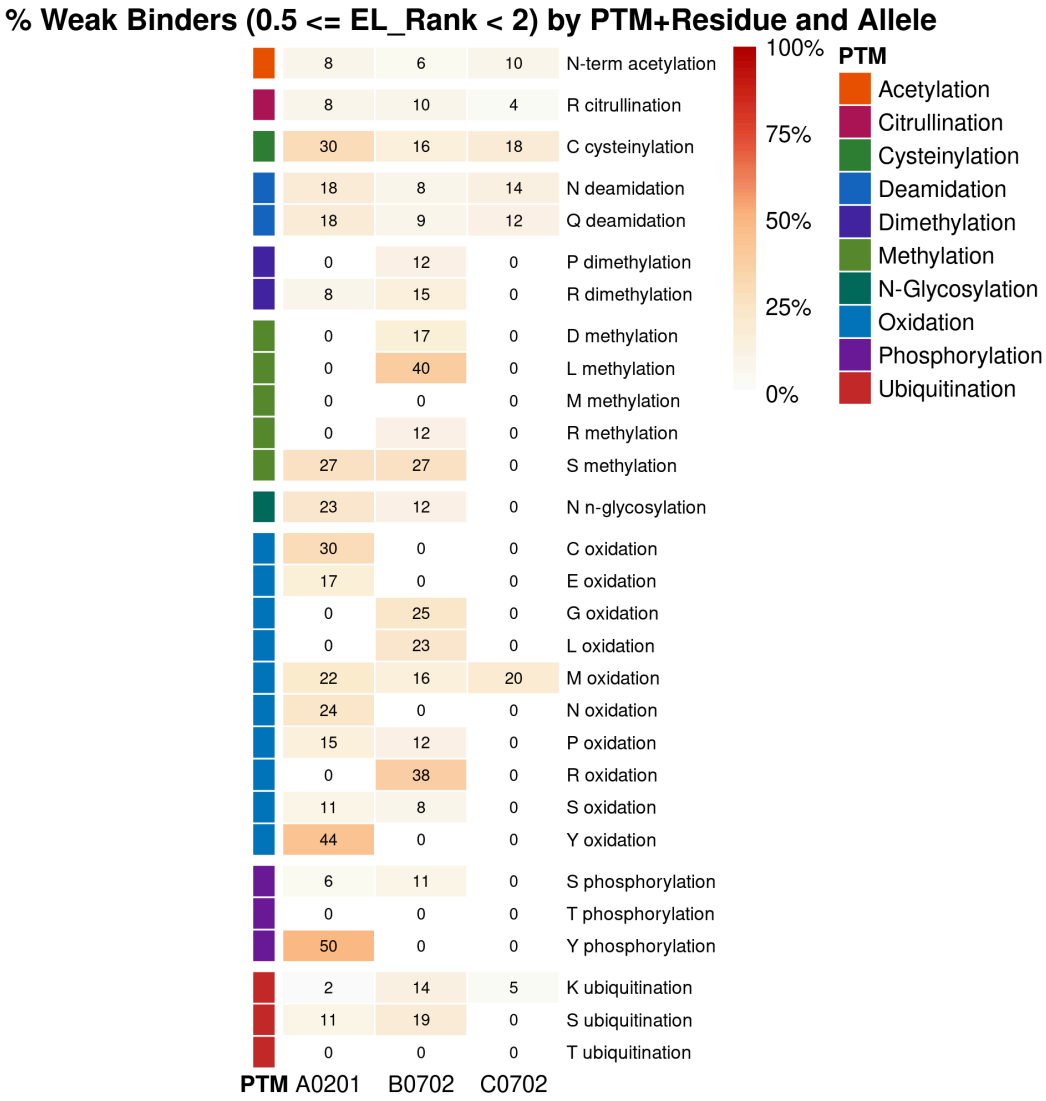
Figure 5A: HLA Binding Affinity

Strong Binders by PTM and Residue



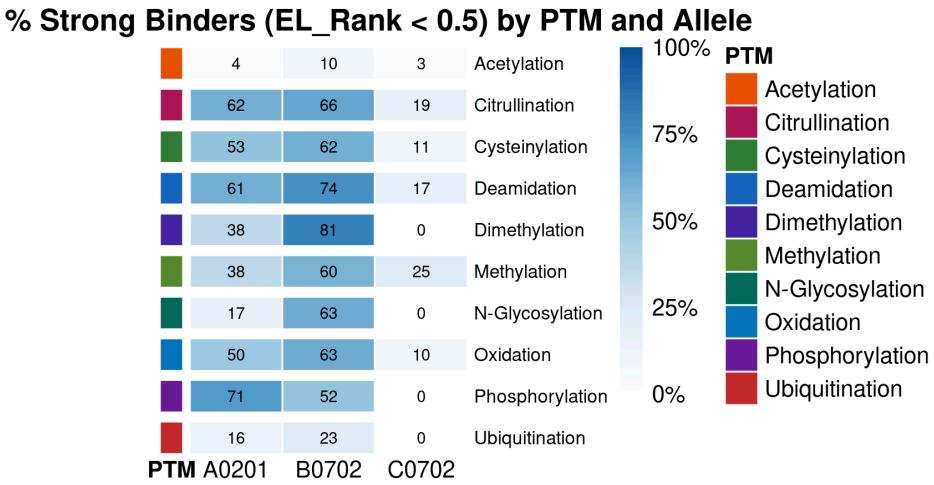
Percentage of strong HLA binders (EL Rank < 0.5%) for each PTM+residue combination across alleles A0201, B0702, and C0702. Combinations with ≥10 peptides.

Weak Binders by PTM and Residue



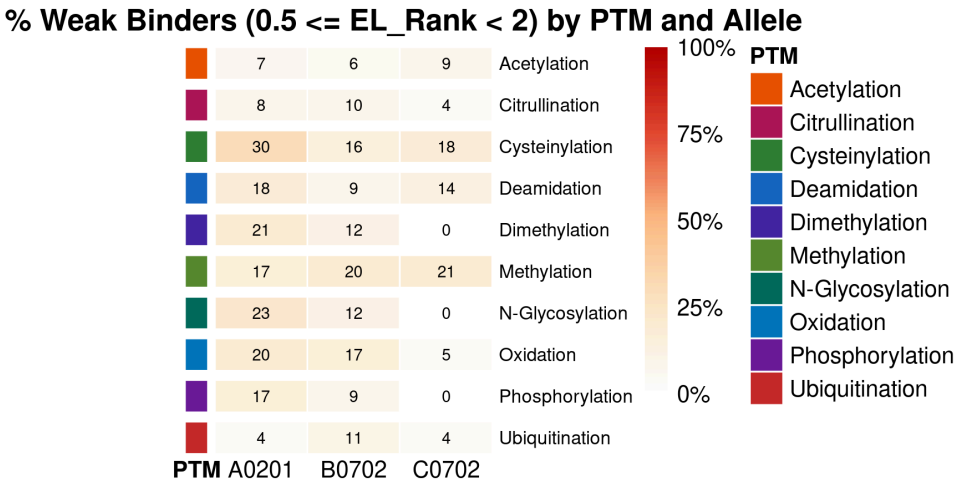
Percentage of weak HLA binders (0.5% ≤ EL Rank < 2%) for each PTM+residue combination.

Strong Binders by PTM Type



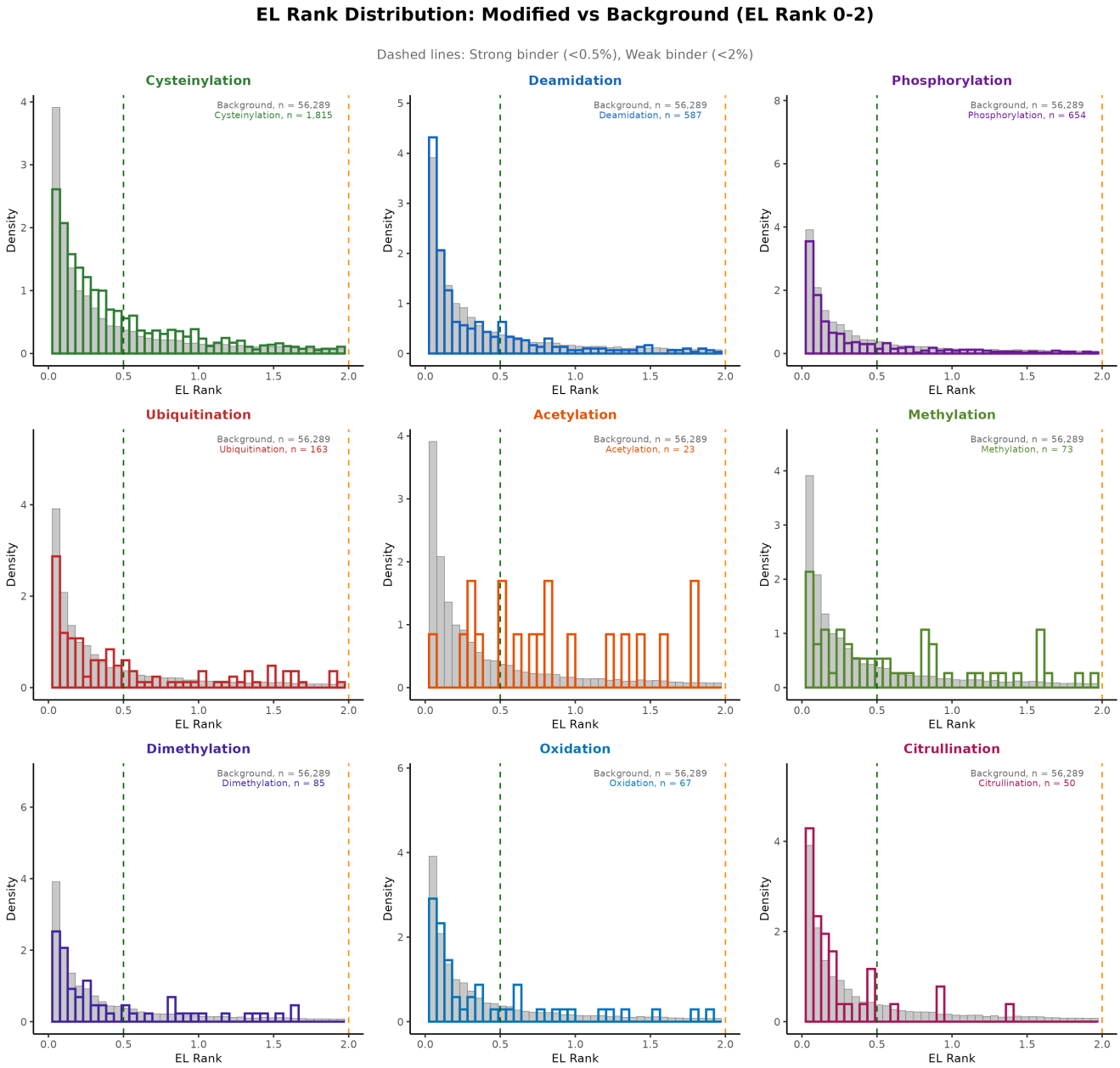
Strong binder percentages aggregated by PTM type across alleles.

Weak Binders by PTM Type



Weak binder percentages aggregated by PTM type.

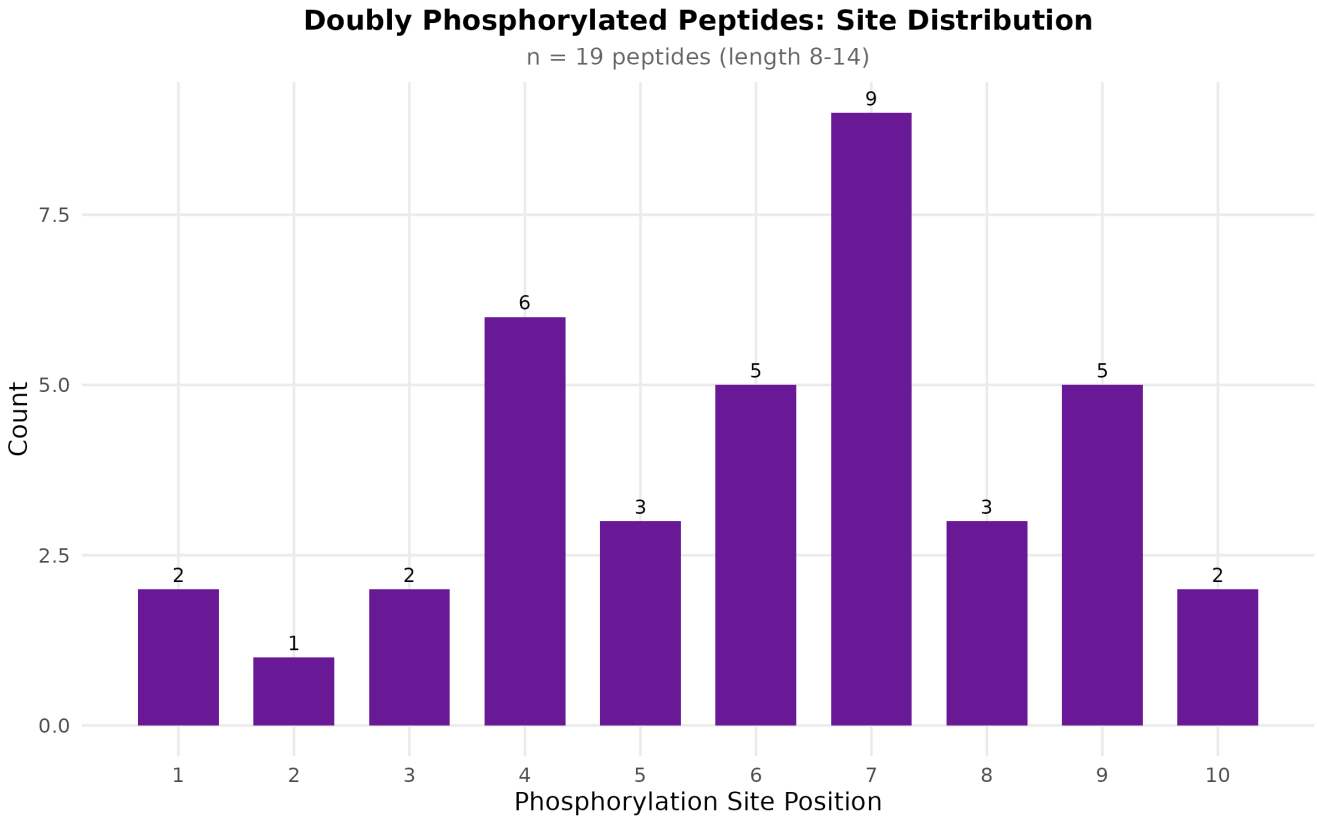
Figure 5B: EL Rank Distribution



EL Rank distribution comparison between modified peptides (colored outlines) and background unmodified peptides (gray) within the binder range (EL Rank 0-2). Dashed lines indicate strong binder (0.5%) and weak binder (2%) thresholds.

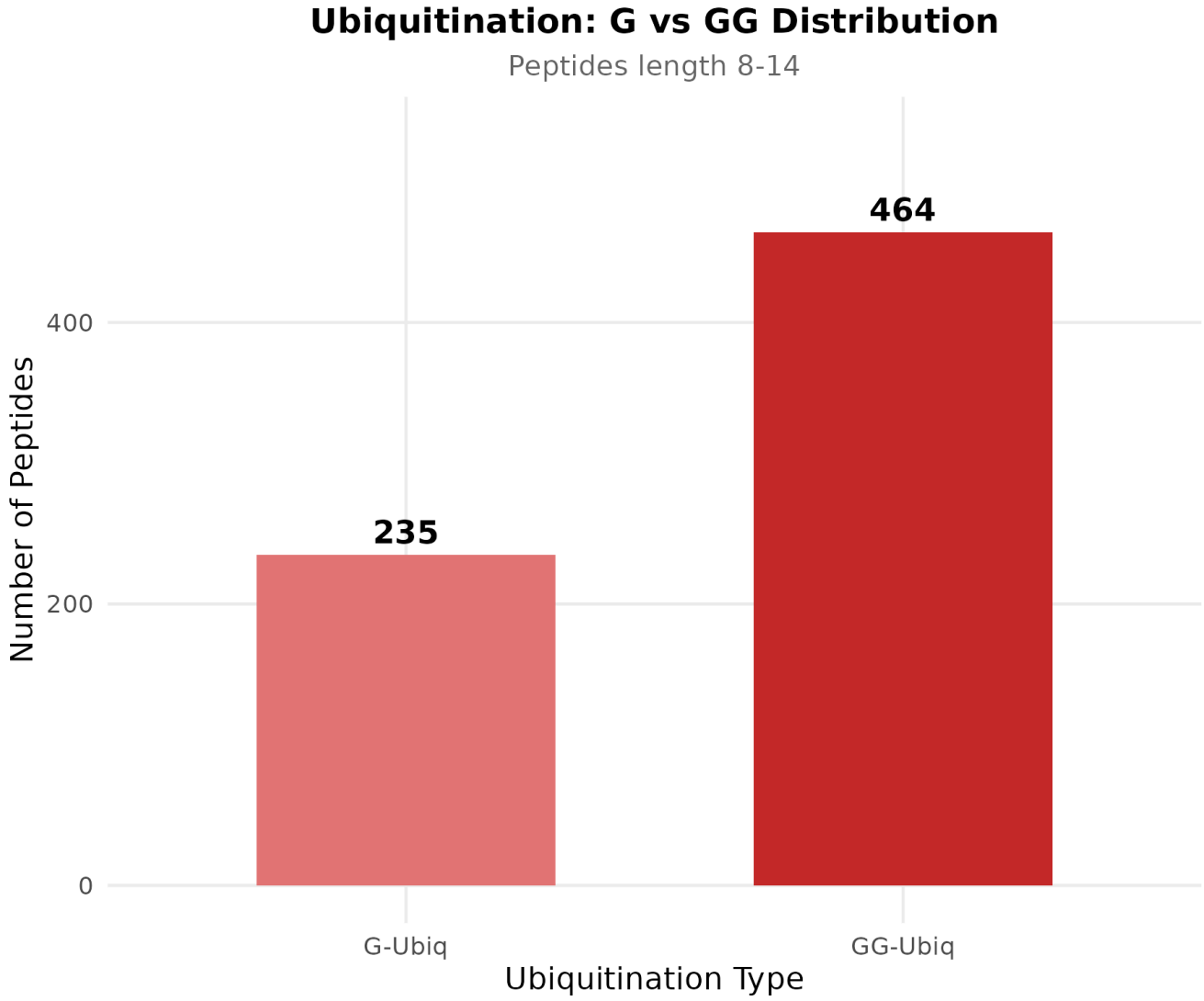
Figure 6: Specific PTM Analyses

Figure 6A: Doubly Phosphorylated Peptides



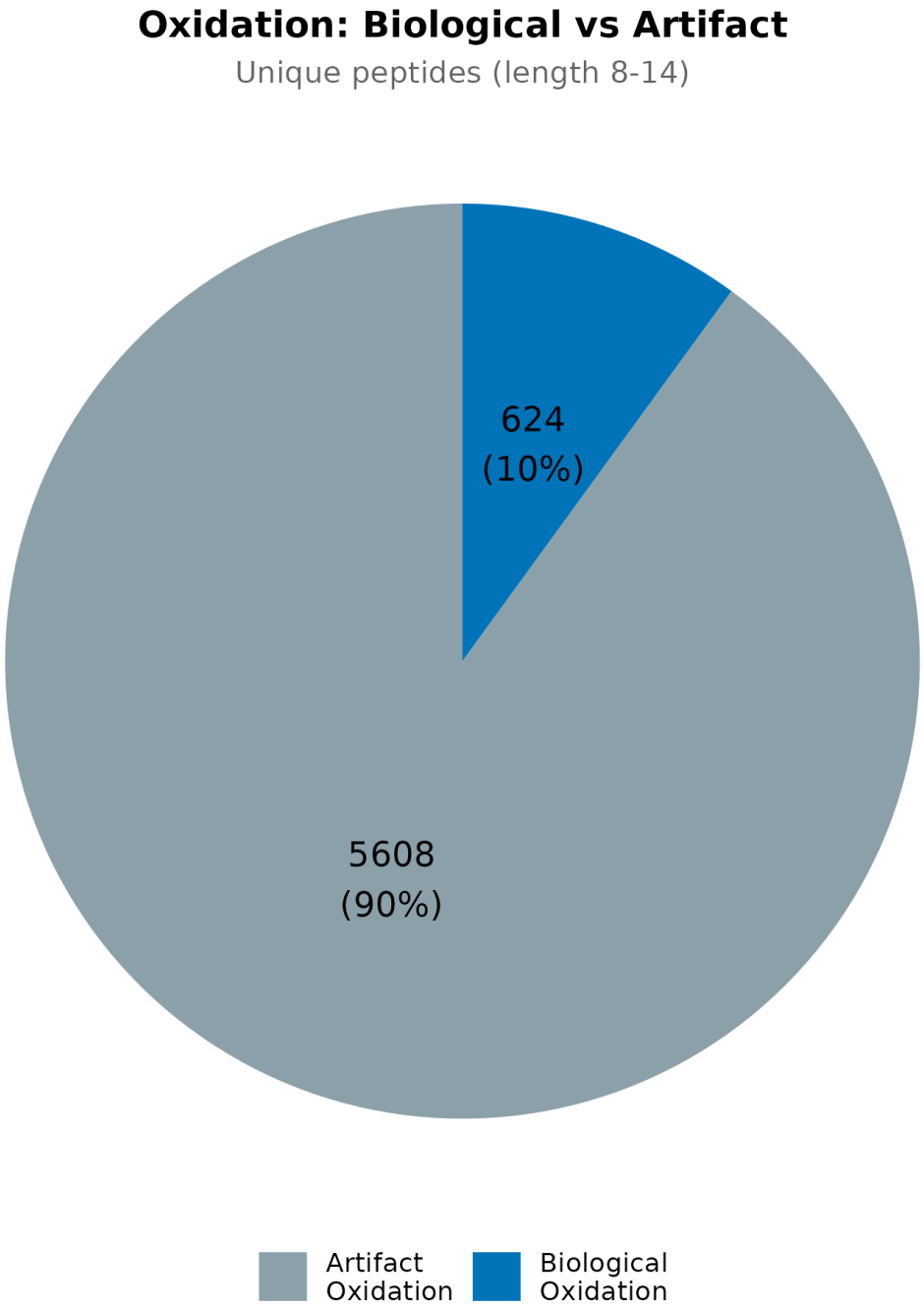
Site distribution for peptides carrying two or more phosphorylation sites.

Figure 6B: Ubiquitination Remnants



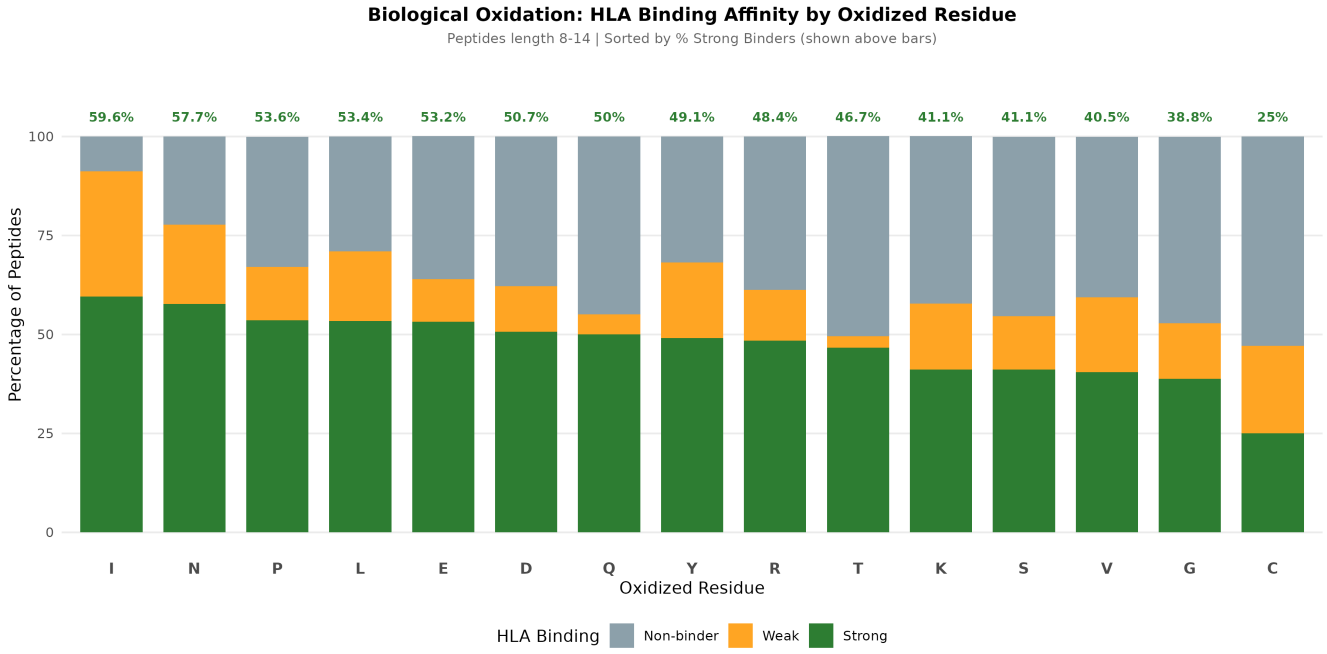
Comparison of ubiquitin remnant types following tryptic digestion: single glycine (G, +57 Da) versus diglycine (GG, +114 Da).

Figure 6C: Oxidation Classification



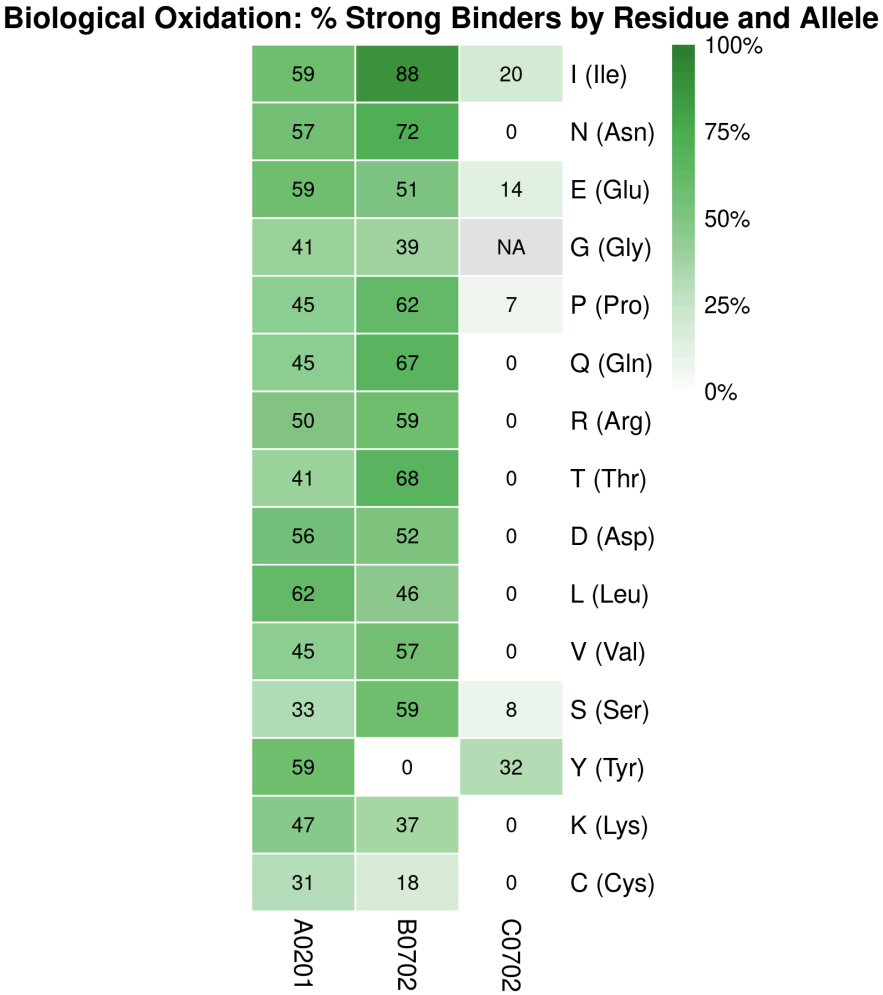
Classification of oxidized peptides: biological oxidation (P, I, L, Q, S, T, V, C, D, E, N, Y, G, K, R) versus artifact oxidation (M, W, H).

Figure 6D: Biological Oxidation by Residue



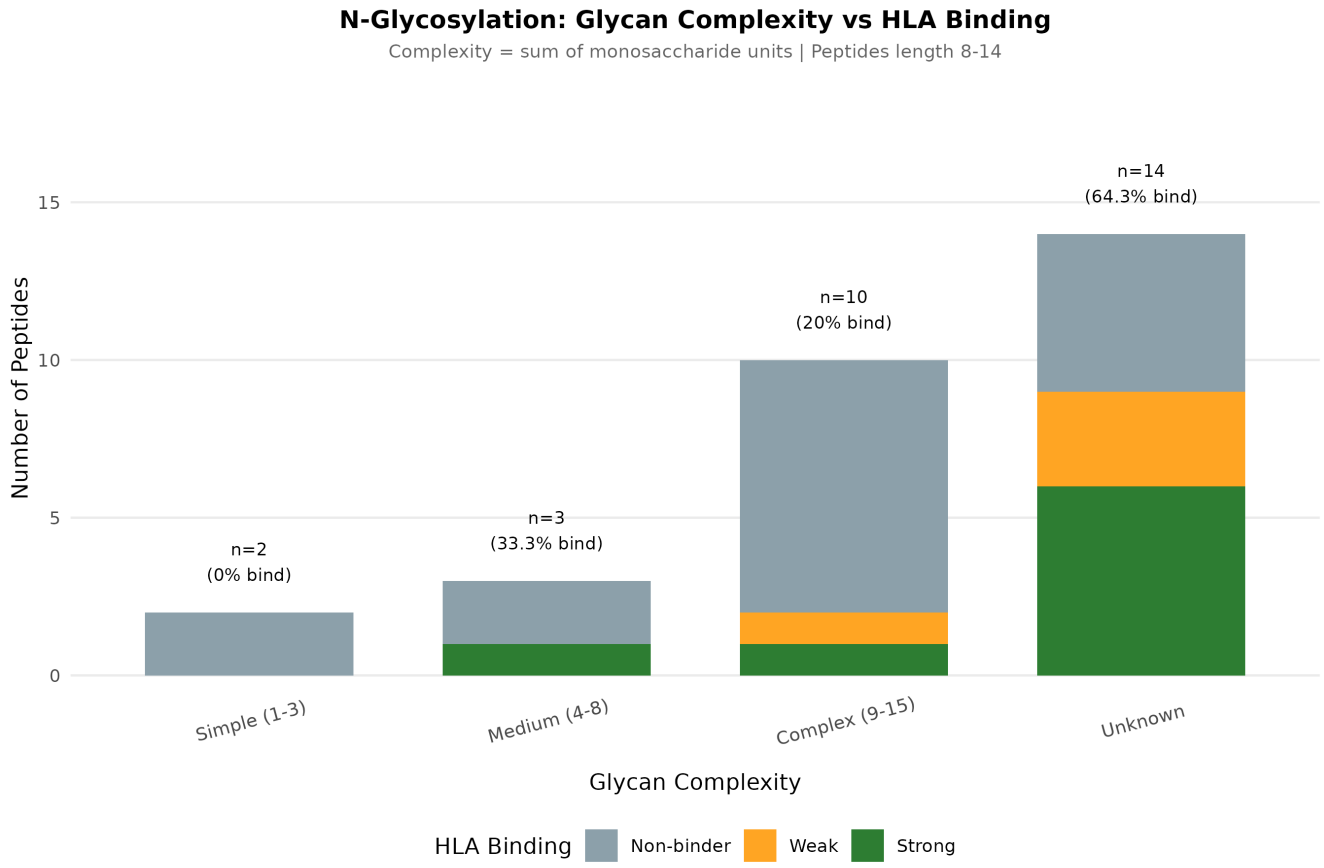
HLA binding classification for biologically oxidized peptides stratified by oxidized residue. Sorted by percentage of strong binders.

Figure 6D2: Biological Oxidation Heatmap



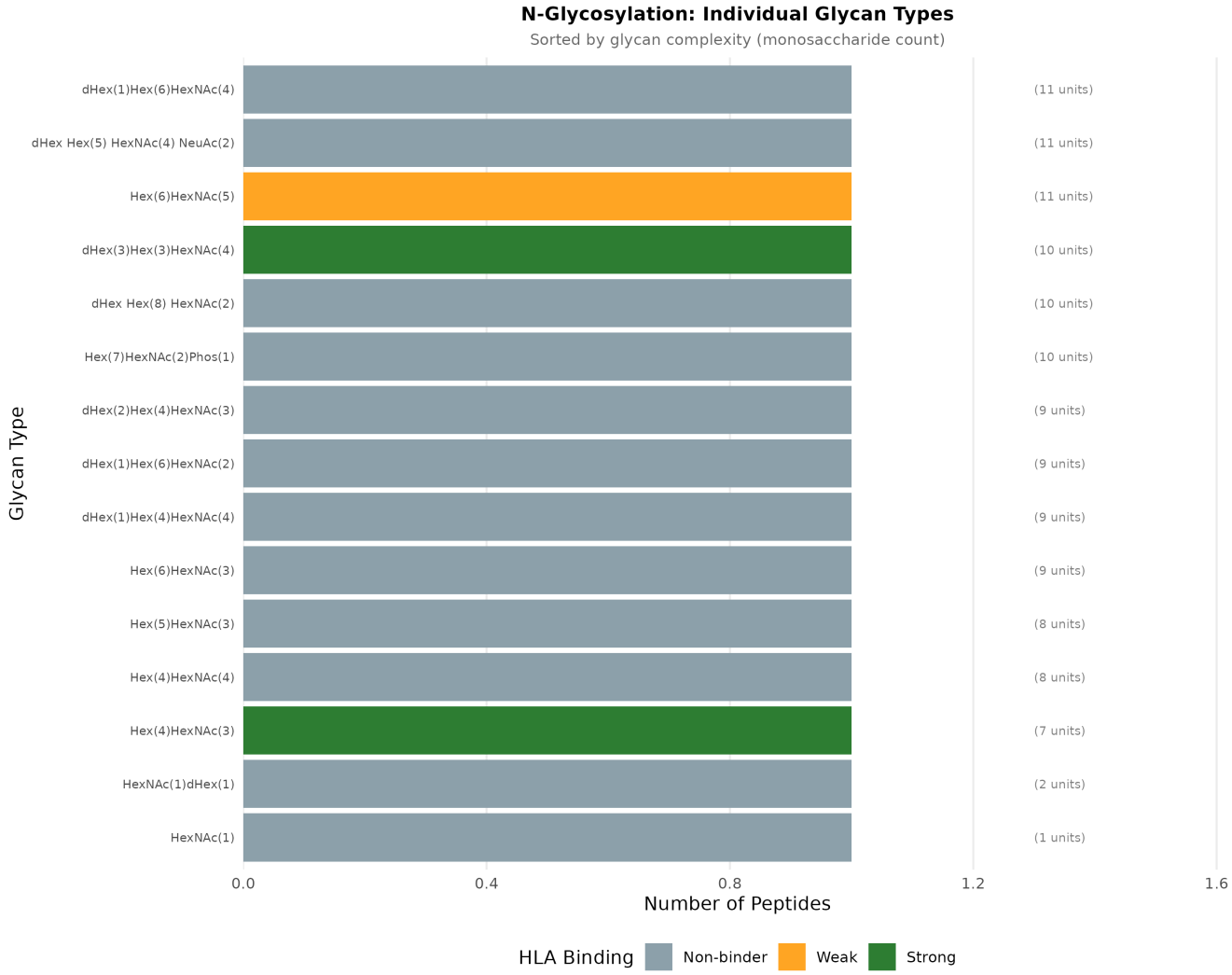
Strong binder percentages for biological oxidation cross-tabulated by residue and HLA allele.

Figure 6E: Glycan Complexity



N-glycosylation analysis stratified by glycan complexity (sum of monosaccharide units). Categories: Simple (1-3), Medium (4-8), Complex (9-15).

Figure 6E2: Glycan Types



Individual glycan structures with HLA binding classification, sorted by monosaccharide count.

Analysis Date: January 2026 | **Made by:** Peter Kubiniok